MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) – REGULAR EXAMINATION

October 2026 – Fifth Semester

SKILL ENHANCEMENT COURSE (SEC)

MG5SECPHYA00

RESEARCH METHODOLOGY, SCIENTIFIC WRITING AND LaTeX FOR PHYSICS

SCHEME OF VALUATION / MARKING SCHEME

Maximum Marks: 50 Duration: 1 hour 30 minutes

Part A: Multiple-Choice Questions (30 marks)

Each correct answer carries 1 mark. No marks for an incorrect or unanswered response.

Q.	Correct answer	Mark
1	B – Statistical techniques and numerical methods	1
2	B – A basis for a research problem	1
3	B – Generate new or unexpected findings	1
4	A – Computational research	1
5	B – Advancement of knowledge	1
6	C – Computational research	1
7	B – Clear, focused and answerable	1
8	D – Defines specific boundaries, scope, population and variables of the study	1
9	A – Closeness of repeated measurements to one another	1
10	D – It cannot be tested	1
11	B – Places the study in the context of existing knowledge	1
12	D – Clearly indicated as quoted text and properly cited with the original source	1
13	C – Findings, evidence, research objectives, and relevant previous studies	1
14	C – The integrity and credibility of research	1
15	D – Words, tables and graphics	1
16	A – The background and research problem	1
17	B – Indented and presented separately	1
18	D – Conducting an extensive literature review	1
19	D – Establish existing knowledge, identify research gaps, and justify the proposed study	1
20	C – The background and context of the research topic	1
21	A – Convert source code into formatted output	1
22	D – Maintains consistency when content changes	1
23	D – Figure environment with `\includegraphics{}`	1
24	B – `\textit{}`	1
25	D – Identify the original source	1
26	D – Organized experimental observations	1
27	D – The structure and alignment of a table	1
28	B – Preparing formal letters and correspondence	1
29	A – Future-oriented language	1
30	D – High-quality typesetting	1
Total		**30**

Part B: Essay Question (20 marks)

Question 31: Prepare a professional Curriculum Vitae (CV) using LaTeX for a final-year undergraduate student applying for a research internship. Include personal/contact information, career objective/profile, educational qualifications, technical/scientific skills, projects/research experience, achieve-

ments/certifications/extracurricular activities, and references/contact details where appropriate. Show suitable LaTeX commands and explain how LaTeX supports consistent professional formatting.

Suggested marking rubric

Assessment criterion	Marks
Document setup and basic structure: `\documentclass`, suitable packages, `\begin{document}` and `\end{document}`	3
Personal/contact details and a suitable career objective/profile	2
Educational qualifications presented clearly and consistently	2
Relevant technical/scientific skills	2
Projects/research experience, with role or outcomes described concisely	3
Achievements, certifications, extracurricular activities and suitable references where appropriate	2
Suitable LaTeX commands/environments for headings and lists (e.g. `\section*`, `itemize`, `enumerate`, or `tabular`)	3
Explanation of consistency, alignment, readability, maintainability and professional presentation	3
Total	**20**

Indicative LaTeX solution

Many valid implementations are possible. Award marks for a logically structured, substantially correct solution; do not require one specific template.

```
\documentclass[a4paper,11pt]{article}
\usepackage[margin=1in]{geometry}
\begin{document}
\begin{center}
{\Large\textbf{Student Name}}\\
Email: student@example.com \quad Phone: 0000000000
\end{center}
\section*{Career Objective}
Seeking a research internship in ...
\section*{Education}
\begin{itemize}
\item B.Sc. Physics, College, Year, CGPA/Marks
\end{itemize}
\section*{Technical and Scientific Skills}
\begin{itemize}
\item Laboratory techniques, data analysis, LaTeX, programming
\end{itemize}
\section*{Projects and Research Experience}
\begin{itemize}
\item Project title -- objective, methods and outcome
\end{itemize}
\section*{Achievements and Activities}
\begin{itemize}
\item Awards, certifications, workshops and activities
\end{itemize}
\section*{References}
Available on request (if appropriate).
```

```
\end{document}
```

Evaluation guidance:

- Accept equivalent commands and alternative professional CV layouts.
- Award partial marks criterion by criterion; do not require one particular package/template.
- Unnecessary sensitive personal details are not required.

Grand Total: Part A 30 + Part B 20 = 50 marks.